

Duplicate synthesis atoms refute the operator-valued Feichtinger conjectures

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Source conjectures

The source paper is K. Mahesh Krishna and P. Sam Johnson, *Operator-Valued p -Approximate Schauder Frames*, arXiv:2201.03955. Conjectures 7.1 and 7.2 appear in Section 7, on page 24 of the source PDF.

Conjecture 7.1. (*Feichtinger conjecture for operator-valued p -ASFs*) *Let $(\{A_n\}_n, \{\Psi_n\}_n)$ be an operator-valued p -ASF in $\mathcal{B}(\mathcal{X}, \mathcal{Y})$ such that*

$$0 < \inf_{n \in \mathbb{N}} \|\Psi_n\| \leq \sup_{n \in \mathbb{N}} \|\Psi_n\| < \infty \quad \text{and} \quad 0 < \inf_{n \in \mathbb{N}} \|A_n\| \leq \sup_{n \in \mathbb{N}} \|A_n\| < \infty.$$

Then $(\{A_n\}_n, \{\Psi_n\}_n)$ can be partitioned into a finite union of operator-valued p -approximate Riesz sequences. Moreover, what is the number of partitions required?

Conjecture 7.2. (*Feichtinger conjecture for operator-valued p -ABSs*) *Let $(\{A_n\}_n, \{\Psi_n\}_n)$ be an operator-valued p -ABS in $\mathcal{B}(\mathcal{X}, \mathcal{Y})$ such that*

$$0 < \inf_{n \in \mathbb{N}} \|\Psi_n\| \leq \sup_{n \in \mathbb{N}} \|\Psi_n\| < \infty \quad \text{and} \quad 0 < \inf_{n \in \mathbb{N}} \|A_n\| \leq \sup_{n \in \mathbb{N}} \|A_n\| < \infty.$$

Then $(\{A_n\}_n, \{\Psi_n\}_n)$ can be partitioned into a finite union of operator-valued p -approximate Riesz sequences. Moreover, what is the number of partitions required?

Definition 7.3. *A operator-valued p -ASF $(\{A_n\}_n, \{\Psi_n\}_n)$ in $\mathcal{B}(\mathcal{X}, \mathcal{Y})$ is said to be p -scalable if there*

Conjecture 7.1 asks whether every operator-valued p -approximate Schauder frame $(\{A_n\}_n, \{\Psi_n\}_n)$ with

$$0 < \inf_n \|\Psi_n\| \leq \sup_n \|\Psi_n\| < \infty, \quad 0 < \inf_n \|A_n\| \leq \sup_n \|A_n\| < \infty$$

can be partitioned into finitely many operator-valued p -approximate Riesz sequences. Conjecture 7.2 states the same assertion for operator-valued p -approximate Bessel sequences.

Relevant source definitions

For Banach spaces X, Y , an operator-valued p -approximate Schauder frame (p -ASF) in $\mathcal{B}(X, Y)$ is a pair $(\{A_n\}_n, \{\Psi_n\}_n)$ with $A_n \in \mathcal{B}(X, Y)$ and $\Psi_n \in \mathcal{B}(Y, X)$ such that the analysis map $\theta_A x = \sum_n L_n A_n x$, the synthesis map $\theta_\Psi z = \sum_n \Psi_n \Gamma_n z$, and the frame operator $S_{A, \Psi} = \theta_\Psi \theta_A = \sum_n \Psi_n A_n$ are well defined and bounded, with $S_{A, \Psi}$ invertible. If invertibility of $S_{A, \Psi}$ is not required, the source calls the pair an operator-valued p -approximate Bessel sequence (p -ABS).

The source defines an operator-valued p -approximate Riesz basis (p -ARB) by the condition

$$P_{A,\Psi} := \theta_A S_{A,\Psi}^{-1} \theta_\Psi = I_{\ell^p(\mathbb{N}) \otimes Y}.$$

A subcollection is an operator-valued p -approximate Riesz sequence if it is a p -ARB on

$$Z = \overline{\text{span}} \bigcup_n \Psi_n(Y).$$

Proof Intuition

The conjectures control only the norms of the individual analysis and synthesis atoms. They do not forbid infinitely many different analysis atoms from sharing exactly the same synthesis atom. In the example below, all bad atoms synthesize into the first coordinate vector of ℓ_1 , while their analysis functionals read all coordinates. Adding the ordinary coordinate atoms for the tail makes the total frame operator invertible, so the pair is a genuine operator-valued 1-ASF. But every finite partition puts two of the duplicate synthesis atoms into the same part, and a Riesz subcollection cannot have non-injective synthesis.

Lemma. *If a subcollection is an operator-valued p -approximate Riesz sequence, then its synthesis operator is injective.*

Proof. For a p -ARB, the source definition gives $I = \theta_A S_{A,\Psi}^{-1} \theta_\Psi$ on the corresponding coefficient space. If $\theta_\Psi z = 0$, then

$$z = \theta_A S_{A,\Psi}^{-1} \theta_\Psi z = 0.$$

Thus θ_Ψ is injective. The same argument applies to a Riesz sequence after restricting to its closed span Z . $\square$

Counterexample. *Conjectures 7.1 and 7.2 of arXiv:2201.03955 are false as stated. There is an operator-valued 1-ASF in $\mathcal{B}(\ell_1, \mathbb{K})$ whose analysis and synthesis atoms all have norm 1, but which cannot be partitioned into finitely many operator-valued 1-approximate Riesz sequences. Consequently it is also a counterexample to the 1-ABS conjecture.*

Proof. Let $X = \ell_1$ over the scalar field $\mathbb{K}$, let $Y = \mathbb{K}$, and write $(e_k)_{k \geq 1}$ for the standard unit vector basis of ℓ_1 . Let e_k^* be the k th coordinate functional, and let $J_k : \mathbb{K} \rightarrow \ell_1$ be $J_k(\alpha) = \alpha e_k$.

Index the family by the countable set

$$I = \{b_k : k \geq 1\} \cup \{g_k : k \geq 2\},$$

identified with $\mathbb{N}$ by any enumeration. Define

$$A_{b_k} = e_k^*, \quad \Psi_{b_k} = J_1 \quad (k \geq 1),$$

and

$$A_{g_k} = e_k^*, \quad \Psi_{g_k} = J_k \quad (k \geq 2).$$

All these operators have norm 1, so the lower and upper norm bounds in Conjectures 7.1 and 7.2 are satisfied.

Since $Y = \mathbb{K}$, the coefficient space $\ell^1(I) \otimes Y$ is canonically $\ell^1(I)$. For $x = (x_k) \in \ell_1$, the analysis sequence is

$$\theta_A x = ((x_k)_{k \geq 1}, (x_k)_{k \geq 2}) \in \ell^1(\{b_k\}) \oplus_1 \ell^1(\{g_k\}),$$

hence

$$\|\theta_A x\|_1 = \sum_{k \geq 1} |x_k| + \sum_{k \geq 2} |x_k| \leq 2\|x\|_1.$$

For coefficients $a = (a_{b_k}, a_{g_k}) \in \ell^1(I)$,

$$\theta_\Psi a = \left(\sum_{k \geq 1} a_{b_k} \right) e_1 + \sum_{k \geq 2} a_{g_k} e_k,$$

so

$$\|\theta_\Psi a\|_1 \leq \sum_{k \geq 1} |a_{b_k}| + \sum_{k \geq 2} |a_{g_k}| = \|a\|_1.$$

Thus the analysis and synthesis maps are bounded.

The frame operator is

$$Sx = \theta_\Psi \theta_A x = \left(\sum_{k \geq 1} x_k \right) e_1 + \sum_{k \geq 2} x_k e_k.$$

It is bounded on ℓ_1 . It is also invertible: for $y = (y_k) \in \ell_1$,

$$S^{-1}y = \left(y_1 - \sum_{k \geq 2} y_k, y_2, y_3, \dots \right),$$

and

$$\|S^{-1}y\|_1 \leq |y_1| + 2 \sum_{k \geq 2} |y_k| \leq 2\|y\|_1.$$

Therefore $(\{A_i\}_{i \in I}, \{\Psi_i\}_{i \in I})$ is an operator-valued 1-ASF in $\mathcal{B}(\ell_1, \mathbb{K})$.

Now take any finite partition $I = I_1 \cup \dots \cup I_m$. The bad set $B = \{b_k : k \geq 1\}$ is infinite, so some part I_j contains two distinct bad indices, say b_r and b_s . In the coefficient space of that subcollection, the vector $\delta_{b_r} - \delta_{b_s}$ is nonzero, but its synthesis is

$$\theta_\Psi^{I_j}(\delta_{b_r} - \delta_{b_s}) = J_1(1) - J_1(1) = 0.$$

Hence the synthesis operator of the I_j subcollection is not injective. By the lemma, that subcollection cannot be an operator-valued 1-approximate Riesz sequence. Since this happens for every finite partition, the frame cannot be partitioned into finitely many operator-valued 1-approximate Riesz sequences.

This disproves Conjecture 7.1. Since every 1-ASF is, after dropping the invertibility requirement, a 1-ABS in the sense of the source definition, the same example also disproves Conjecture 7.2. $\square$

Verification notes

The construction uses only the source paper's definitions. The key check is that the duplicated synthesis maps do not prevent boundedness of the synthesis operator when $p = 1$: summation of the bad coefficients is a bounded functional on ℓ^1 . The added good coordinate atoms make the total frame operator invertible by an explicit bounded inverse.

No numerical computation is needed. The crop was generated by `code/make_open_problem_crop.py` from the local source PDF.

Novelty check and limitations

A bounded novelty check was performed on 2026-06-19. The run indexes were searched for arXiv:2201.03955 and the core phrases “operator-valued p -approximate Schauder frame”, “operator-valued p -ASF”, “Feichtinger conjecture”, and “ p -approximate Riesz sequence”. External web/arXiv queries for the exact title and for combinations of “operator-valued p -approximate Schauder frame”, “Feichtinger conjecture”, and “counterexample” found the source paper and adjacent frame-conjecture papers, but no direct counterexample to Conjectures 7.1–7.2.

The counterexample is for $p = 1$. It refutes the conjectures as written for arbitrary $p \in [1, \infty)$, but it does not address possible repaired statements for fixed $p > 1$, for Hilbert/ G -frame subcases, or under hypotheses that prevent repeated synthesis atoms.

Human-review recommendation

Review as a likely valid counterexample to Conjectures 7.1 and 7.2 as stated. The main points to check are the interpretation of a Riesz subcollection on the closed span Z and the injectivity consequence of the source definition $P_{A,\Psi} = I$.

References

- [1] K. Mahesh Krishna and P. Sam Johnson, *Operator-Valued p -Approximate Schauder Frames*, arXiv:2201.03955, 2022.